# WordPress libheif RCE

**WordPress libheif RCE** - Adrian Tiron, Fortbridge.


- Published: 2026-10-05
- Original: <https://fortbridge.co.uk/research/wordpress-libheif-rce/>
- Preserved from: https://fortbridge.co.uk/research/wordpress-libheif-rce/ (manual-import) on 2026-10-21
- Licence: unknown



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## Content

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WordPress libheif RCE: Exploit Chain

[image: WordPress libheif RCE chain on Ubuntu and Debian]

## Motivation

Hacktron's HEIF research was genuinely inspiring. It showed what can happen when a memory corruption bug is reachable through an ordinary web application, but the exploit details were not published. We wanted to understand that missing middle for ourselves.

This follows our earlier libheif research in a Next.js sharp stack, with its reproducible exploit repository. Here we examine a different libheif flaw and a WordPress/ImageMagick/PHP-FPM route.

This was also a chance to bring two security disciplines together. The entry point looks like normal web testing, an authenticated image upload to WordPress. Exploiting it requires image-format construction, fuzzing, GDB, allocator analysis and C++ object layout work. The OWASP Top 10 remains useful, but it does not cover the native attack surface behind modern web applications.

We first built the exploit for an Ubuntu setup. We then repeated the work against Debian, with different WordPress, PHP-FPM, ImageMagick, libheif and glibc packages. That second profile made the proof of concept more robust and gave us a useful cross-check of the original assumptions.

## TL;DR

This WordPress libheif RCE chain supports two exact native stacks. An authenticated WordPress Author uploads HEIC images through the normal Media Library API. WordPress-generated image derivatives disclose the live library and heap addresses needed by the final payload. A profile-specific `unci` image then turns a file-controlled heap overflow into a hijacked C++ virtual call and executes `id` as the PHP-FPM account.

The proof is not a worker crash or an HTTP 503. Before each run, the exploit verifies that the profile's proof path returns 404. It reports success only when a later GET of that path returns HTTP 200 with:

```
uid=33(www-data) gid=33(www-data) groups=33(www-data)
```

Before building the malicious image, the exploit uploads separate disclosure images and examines the derivative JPEGs returned by WordPress. Values reconstructed from their pixels reveal library addresses and identify which of our two exact tested native builds is running; the exploit then chooses that build's measured profile.

Unknown, incomplete or conflicting address evidence stops the run before a trigger is built. In our final validation, the Ubuntu profile succeeded for 6 of 8 fresh PHP-FPM parent processes, and the Debian profile succeeded for 22 of 24.

## Scope and tested profiles

The tested threat model is an authenticated WordPress account with the `upload_files` capability. In a standard WordPress installation, that normally means Author or higher. A plugin that exposes the same image-processing path to guest uploads may lower that requirement, but we have not claimed or tested that case here.

The lab uses an eight-child dynamic PHP-FPM pool. This matters because the exploit occupies seven workers with slow image requests and reserves the remaining worker for address recovery, calibration and the final trigger. A different pool size or process-manager policy needs a separately tested scheduling profile.

- **Ubuntu 26.04:** WordPress 7.1.1, PHP-FPM 8.5.4, ImageMagick 7.1.2.18, libheif 1.21.2 and glibc 2.43 on amd64.
- **Debian 13:** WordPress 7.0, PHP-FPM 8.4.24, ImageMagick 7.1.1.43, libheif 1.19.8 and glibc 2.41 on amd64.

*The two validated native stacks.*

## The file-controlled heap overflow

GHSA-x8r2-mggj-j6wr affects libheif's ISO/IEC 23001-17 uncompressed image decoder. In the mixed-interleave YCbCr path, a file can declare Cb as 16-bit and Cr as 8-bit. libheif allocates the Cr plane for one byte per sample, but the paired chroma loop uses Cb's two-byte width for both destinations. Each Cr write therefore advances too far, eventually carrying file-controlled bytes beyond the Cr allocation.

An image plane is the memory buffer and metadata for one colour channel. libheif stores references to those planes in a C++ channel vector. On useful heap layouts, the overflowing Cr plane is close to that vector or a decoder object. Changing the vector can redirect later plane writes. Changing the decoder's first machine word controls its virtual-table pointer, usually shortened to `vptr`.

A C++ virtual call reads a function address from the object's virtual table. Decoder teardown makes such a call. Once the exploit controls the `vptr`, it can choose the later control-flow target, provided it also knows the current library and heap addresses.

*The crafted image controls the decoder `vptr`.*

That control is only one part of the chain. ASLR moves libc and ImageMagick whenever a new PHP-FPM parent process starts. Useful heap objects also move between workers. The exploit must identify the exact native build and recover current addresses before it can construct a trigger.

## Recovering addresses from returned pixels

The disclosure stage uses the derived-item and plane-geometry flaws described by GHSA-2jg2-4ch7-h545. An identity-derived image can claim dimensions larger than its backing plane. A later crop copies beyond the real allocation and carries neighbouring heap bytes into the decoded output.

WordPress writes its derivative as a lossy JPEG, so the client cannot read raw leaked bytes directly. Our disclosure image moves the over-read data into luma and expands each source byte into a uniform 8 by 8 block. The exploit takes the median value of each returned block. It then validates complete 64-bit pointer relationships instead of trusting isolated values that happen to look like addresses.

The returned pixels provide four useful classes of evidence:

- glibc arena pointers, which resolve to one page-aligned libc base;
- thread-arena and decoder pointers used by the Ubuntu heap strategy;
- exact native-library pointer pairs used to identify the installed build; and
- on Debian, two adjacent MagickCore callback pointers that resolve its live module base.

*Live addresses recovered from returned pixels.*

## Selecting the exact build

The glibc arena pointers are not a unique operating-system fingerprint. Different public libc builds can share the same spacing and page offset. A server banner is also too weak to choose memory-layout constants.

The selector requires a complete native-module pattern from returned pixels. Ubuntu uses an exact pair of pointers inside its libheif artifact. Debian can use a libheif pair, then confirms the adjacent `CompareSplayTreeString` and `RelinquishMagickMemory` pointers inside

its MagickCore artifact. All pointers must resolve to one declared build and one page-aligned module base.

The selected native stack is bound to one libc profile. The exploit also requires one unambiguous libc base. A partial pair, an unknown pair, mixed-build evidence or competing bases stops the run. The HTTP `Server` header can choose which harmless allocation schedule to try first, but it never decides the profile.

## Reserving one PHP-FPM worker

Connection reuse does not guarantee that Apache will select the same PHP-FPM worker. The exploit instead sends seven slow, valid HEIC uploads to occupy seven children. A shorter gate request finishes first. The remaining worker then handles address recovery, heap calibration and the trigger.

Every child created by one PHP-FPM parent process inherits the same library layout. Individual workers can still have different glibc thread arenas. Library bases can therefore be reused while the same parent process remains alive, but heap-specific calibration must match the worker that builds the final decoder.

## Ubuntu control-flow route

The Ubuntu trigger is a 1024 by 128 mixed-interleave image. Its controlled Cr tail crosses three small allocations and reaches the live channel vector. The overflow replaces one plane pointer in that vector. Later chroma rows then write a compact fake dispatch structure into the decoder address predicted by the thread-arena leak.

The decoder’s overwritten `vptr` points slot 1 at a dispatcher inside the exact Ubuntu libc build. The dispatch structure contains an alignment-safe `do_system` call site and a pointer to the short command. The live libc base comes from returned pixels. The dispatcher and call-site offsets are hardcoded build constants taken from the verified public package.

## Debian control-flow route

The Debian profile needs a different layout. Its trigger is 512 pixels wide, and source row 32 reaches the end of the Cr allocation. The overflow must change the pointers in libheif’s channel vector without breaking the next writes.

We observed two alignments of those pointer fields in the tested worker layout. The crafted bytes spanning rows 32 and 33 restore the displaced vector metadata and complete the Cb and Cr destination pointers in either observed alignment.

Subsequent decoder writes can then carry the fake virtual table and command into a writable data region of ImageMagick’s MagickCore library. RELRO makes some library regions read-only after loading; this selected data region remains writable. The result is specific to the measured Debian build and these two layouts.

A separate 64-frame HEIC acts as a leak-preparation image. It creates ordinary MagickCore objects that the following disclosure request may reuse. The returned pixels can then expose adjacent `CompareSplayTreeString` and `RelinquishMagickMemory` callback pointers. Their hardcoded offsets, `+0x1c2f90` and `+0x13ca40` in this exact artifact, must agree on one canonical, page-aligned MagickCore base.

The final fake virtual table, command object and dispatcher slot are written only inside the profile's declared `+0x2c7000..+0x3038b8` writable interval. If a previous trigger replaced the reserved worker, the next position can reuse the process-wide libc and MagickCore bases. Otherwise, a dedicated HEIC request refreshes that worker before the next heap attempt. Cached bases are discarded when the PHP-FPM parent process changes.

## Runtime addresses and hardcoded build constants

The exploit keeps two kinds of value separate:

- **Runtime addresses**, including libc, MagickCore, thread arenas and decoder anchors. These are recovered from returned pixels for every fresh PHP-FPM parent process.
- **Hardcoded build constants**, including symbol offsets, instruction signatures, object-field offsets, trigger geometry and permitted writable intervals. These are measured offline from exact public binaries and stored in `profiles/build_profiles.json`.

Fingerprinting the operating system is not enough. A profile identifies the complete native stack, including PHP-FPM, php-imagick, ImageMagick's HEIC coder, MagickCore, libheif, libc and libstdc++. A package update can change pointer offsets, object layouts or usable code locations. A different set of artifacts therefore needs a new measured and verified profile.

## End-to-end request flow

```
1. Authenticate to WordPress and obtain a REST nonce
2. Upload disclosure images and download generated derivatives
3. Validate one exact native stack and one libc base
4. Select the matching scheduling and heap strategy
5. Build an ASLR-adjusted unci trigger from returned addresses
6. Upload the trigger through the WordPress media route
7. Fetch the previously absent proof path and require uid= output
```

The trigger upload and the proof check are different HTTP requests. The upload can end in HTTP 503 when the affected PHP-FPM child crashes or exits before completing its response; that only tells us the worker failed, not whether the command ran.

Before the trigger, the exploit checks that its selected proof URL returns 404 (the Debian profile uses `/rce-proof.txt`). If the command executes, it writes the output of `id` to that URL's web-accessible path.

We count execution only when a later, separate GET of the same URL returns HTTP 200 containing the expected `uid=33(www-data)` output. A 503 alone is crash telemetry, never RCE proof.

## Reliability and validation

A worker is one PHP-FPM child. For each independent validation run, we restart PHP-FPM so it has a new parent process and children with newly randomised address layouts. We then try no more than the stated number of candidate heap positions before scoring that run.

Attempts within one run are related because its workers inherit the same parent process's library layout; the number of uploads is not the number of independent trials.

### Independent validation results

Ubuntu succeeded for **6 of 8 fresh parent processes, 75 percent**, within a limit of 8 candidate positions. Across that validation set, the exploit tried 26 positions. Six produced the `id` proof, 19 fired without a proof, and one was rejected safely before the trigger. Profile selection made 142 separate disclosure uploads and safely rejected 134 incomplete responses before selecting one exact profile for each run.

Debian succeeded for **22 of 24 fresh parent processes, 91.7 percent**, within a limit of 16 candidate positions. Across that validation set, the exploit tried 93 positions. Twenty-two produced the proof, 33 fired without a proof, and 38 were rejected safely before the trigger. Profile selection made 78 disclosure uploads and safely rejected 54 incomplete responses.

These categories matter. A classification rejection means the exploit does not yet have enough evidence to choose a build. A safe pre-trigger rejection means the build is known, but the returned heap state fails a structural check. A fired miss means a trigger was uploaded, but the proof path remained absent. An execution means the independent request returned `uid=33(www-data)`.

The results are strong for the two exact lab profiles, but they are not universal success rates for WordPress or libheif. Heap placement depends on the process state, pool settings and traffic. Package changes require a new profile. The repository includes the reduced evidence, 70 regression tests, exact lab containers and the code used for these measurements.

*The exploit returns the `id` proof.*

## Detection and mitigation

- **Patch libheif.** GHSA-x8r2-mggj-j6wr affects versions 1.18.0 through 1.23.2 and is patched in 1.23.3. GHSA-2jg2-4ch7-h545 is patched in 1.23.2. Use a current distribution security package and verify the library loaded by the image stack.
- **Reduce the decode surface.** If HEIC and AVIF uploads are unnecessary, reject them before native decoding. A custom libheif build can also omit the uncompressed codec.

- **Isolate image processing.** Decode untrusted media in a disposable, least-privileged service with restricted network access, a read-only application tree and no application secrets.
- **Treat native crashes as security events.** Correlate repeated FPM child exits and 503 responses with uploads containing `unci`, `iden`, crop, overlay or grid relationships.
- **Constrain impact.** Restrict writable paths and prevent script execution from upload directories. This does not fix memory corruption, but it limits useful post-exploitation destinations.

## Credits

Alex Thomas and Wordfence deserve primary credit for finding, validating and coordinating the mixed-interleave issue. The advisory reports manually validated file disclosure and command execution on a specific WordPress deployment, alongside a public crash reproducer. Its deployment demonstration used target-specific runtime library bases and did not publish the final RCE payloads.

The libheif maintainers and the researchers credited in the advisories documented the affected code and fixes. Hacktron's public HEIF work motivated us to investigate how native image bugs can be exploited through web applications.

Our contribution is a publicly released, reproducible end-to-end WordPress exploit for this specific advisory that recovers the required runtime addresses from HTTP-returned image pixels rather than supplying target process mappings.

It includes exact Ubuntu and Debian build profiles, PHP-FPM scheduling and bounded validation results. To our knowledge, this is the first public exploit chain for this issue with that complete remotely calibrated path; it is not a claim to have discovered the vulnerability.

## Proof-of-concept code

**The exploit, payload generators, build profiles, reproducible Ubuntu and Debian containers, tests and reduced validation evidence are available in the Fortbridge WordPress libheif RCE repository.**

## References

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